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Review

Return to Play Following Achilles Tendon Rupture in NFL Players Based on Position

Dane Barton, MD,
Aditya Manoharan, MD,
Ansab Khwaja, MD,
Jacob Sorenson, and Michel Taylor, MD

Abstract: Background: *The purpose of this study was to determine the return-to-play (RTP) rate and postinjury performance after Achilles tendon (AT) ruptures in National Football League (NFL) skill position players.* Methods: *The study included NFL skill positions with an AT rupture between the 2009-2010 and 2015-2016 seasons. Performance data were collected and compared against a matched control group. RTP was defined as playing in at least 1 game after repair.* Results: *RTP rate was 57% for the study cohort. The tight ends (TEs) had the highest RTP rate at 71% while the wide receivers (WRs) had the lowest RTP rate at 38%. Compared with the control group, WRs with successful RTP had significantly less receptions per game (P = .01). For defensive players with RTP there were significant decreases in postrepair performance in tackles, passes defended, and fumbles forced/recovered compared with the control group.* Conclusion: *A total of 57% of players achieved RTP with WRs and running backs (RBs) having the lowest RTP rates and TEs and linebackers (LBs) having the highest RTP rates. RBs, defensive backs (DBs), and LBs with successful RTP had decreased performance in all categories. This updated information may be helpful for athletes, physicians, scouts, and coaches in evaluating players with a history of AT rupture.*

Levels of Evidence: *Analytic, level 3, retrospective cohort study, Epidemiologic study*

Keywords: Achilles tendon rupture; rupture; NFL; return to play

Achilles tendon (AT) ruptures are common and potentially devastating injuries.[1] The incidence of AT rupture is increasing in the general population[2] with rates reported from 7 to 40 per 100 000 person-years.[3-6] Athletic activities account for 68% of all AT ruptures.[7] The AT is the strongest tendon in the human body.[8,9] It is formed by the confluence of the soleus and gastrocnemius muscle tendons and functions as a crucial element of the plantarflexion mechanism. The most common location for tendon rupture is in the midsubstance at the hypovascular or watershed zone, approximately 2 to 6 cm proximal to the insertion on the calcaneus.[8,10] Some have postulated that underpronation increases stress on the AT near the watershed zone.[11] Modifiable risk factors for AT rupture include periodic athletes or "weekend warriors," use of fluoroquinolone antibiotics, oral corticosteroids, and local steroid injections.[5,12,13] AT rupture typically occurs with a sudden forced plantar flexion moment or a violent dorsiflexion force on a plantar flexed foot.[10] While the full mechanism of tendon failure is not fully understood, intrinsic and extrinsic factors are involved, including pathologic degeneration, tendon hypovascularity, and rapid acceleration and deceleration.

Treatment options for acute AT rupture include both operative and nonoperative treatment and there is much debate as to which option is preferable. Nonoperative treatment is comprised of early immobilization and weightbearing restrictions followed by supervised physical therapy including accelerated functional rehabilitation.[14] Historically, nonoperative management was associated with increased rerupture rates[15] and decreased tendon strength;

DOI: 10.1177/19386400211040351. From Department of Orthopaedic Surgery, University of Arizona, Tucson, Arizona (DB, AM, AK); University of Arizona School of Medicine, Tucson, Arizona (JS); and Hughton Clinic, Nashville, Tennessee (MT). Address correspondence to: Dane Barton, MD, Department of Orthopaedic Surgery, University of Arizona, 1501 North Campbell Avenue, 8th floor, 8401, Tucson, AZ 85724; e-mail: dbarton@ortho.arizona.edu

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however, recent studies have challenged these findings.[14,16] Operative management for acute AT rupture includes open end-to-end repair or mini-open or percutaneous repair.[5] The most feared factor of operative treatment is wound complication with an incidence of 5% to 7%.[17]

Previous reports studying acute AT ruptures in high-level athletes have shown return-to-play (RTP) rates from 70% to 82% in professional soccer players[18,19] as well as a recent report of 92% RTP for college football players.[20]

Acute AT injuries, while unfortunate in the general population, are potentially disastrous and career ending for the competitive athlete.[21,22] In the National Football League (NFL), a player who sustains an acute AT rupture is most often treated with primary operative repair and a period of immobilization. This is followed by an often lengthy period of rehabilitation before the athlete is able to RTP.[23] Studies looking at RTP rates following acute AT ruptures and the injury's eventual effect on athletic performance are helpful to team physicians when treating and counseling athletes as well as to coaches, scouts, and general managers who are tasked with making roster personnel decisions.

Previous studies looking at RTP rate following AT rupture in the NFL athlete based on position found RTP rates between 61% and 100% depending on position played.[22,24-26] The quarterback position had a reported 100% RTP rate.[24] Each of these reports divided NFL players by position group and reported an RTP rate for the overall cohort as well as for each position group. Some reports have examined the performance of NFL players after AT rupture; however, those studies used a power ranking system[27] or substituted the number of games played and started[22] as a surrogate for in game performance metrics. Additionally, some reports, while reporting on AT rupture in the NFL, combined NFL data with other professional sports or compared a host of orthopaedic injuries in the NFL and did not focus solely on acute AT rupture.[25,26] While providing supporting evidence to the available literature on the RTP rates of NFL players with AT injuries, we sought to provide detailed comparison performance data for specific positions both before and after injury.

The purpose of this study was to determine the RTP rate after acute AT rupture and primary operative repair of offensive and defensive skill positions in the NFL. We included wide receivers (WRs), tight ends (TEs), running backs (RBs), defensive backs (DBs), and linebackers (LBs). Additionally, we examined changes in player performance after AT rupture and repair by comparing pre- and postinjury individual performance statistics.

Methods

A record of NFL injury reports spanning the 7 seasons between 2009-2010 and 2015-2016 seasons was compiled from publicly available data. Sources included official press releases, team websites, NFL.com, and pro-football-reference.com. Injuries were tracked during the preseason, regular season, playoffs, and offseason. The list was initially filtered for injuries involving the foot and ankle. The list was further narrowed by offensive and defensive skill positions, including WR, RB, TE, DB, and LB. We did not include offensive line, defensive line, or quarterback positions in this study. Each remaining injury was then researched individually in publicly available press releases, NFL websites and injury reports to determine if an AT injury and subsequent repair occurred. If a player was found to have sustained an AT rupture and repair, a more in-depth search was performed to determine if he had experienced prior AT rupture. Players with prior AT injuries were excluded from the study. If a player had not recorded any game statistics prior to AT rupture that occurred during the study period, this player was counted in the RTP group but was not included in the analysis with the control players. Information regarding concomitant injuries was not readily available and thus not recorded. A total of 63 players were identified and met criteria to be included in the study.

For each player it was determined whether they achieved successful RTP after AT repair. Successful RTP was defined as playing in at least one NFL game following the injury.[22,24-26,28] We also calculated the time to RTP and the average time to RTP based on position groups.

Individual performance statistics were recorded for each player included in the study. For both offensive and defensive positions the total number of career games played, percent games started, age at time of injury, timing of injury, RTP, and time to RTP were noted. For the WR and TE groups, yards per reception, receptions per game, and receiving touchdowns (TDs) per season were recorded. For the RB group, yards per rush, rushes per game, and rushing TDs per season were recorded. For both the DB and LB groups the following performance metrics per season were recorded: tackles, tackles for loss, sacks, passes defended, interceptions, fumbles forced and recovered, and total defensive TDs. Cornerbacks and safeties were combined in the DB group.

A matched control group was identified for those players who achieved RTP; performance statistics were analyzed between these groups. An algorithm was designed to identify matched controls for the RTP group. We identified a player of the same position from the same team roster who had the same consecutive healthy seasons to match the injured player, such that both position groups were 1:1 with the control group. The healthy seasons included the 2 seasons preceding and the 2 seasons following the injury. A total of 4 years of performance data were collected for the injured players and the control players. The injury season was not included in either group. A healthy season was defined as playing in a minimum of 12 games. If no player on the same team roster met the inclusion criteria, a player was selected from a team roster in the same division. The exclusion criteria for the control group included any history of AT injury, a foot/ankle injury, or any injury that resulted in the player missing more than 4 games during any season

during the study period. The performance data for the control players were gathered in a manner similar to the study group. In total, 27 players comprised the control group. Some control players were used twice if they were the closest match to an injured player.

Post hoc analysis was completed using a paired *t* test for pre- versus postinjury and RTP statistics. An unpaired *t* test was used to compare statistics for preinjury players who achieved RTP versus those who did not achieve RTP. Additionally, an unpaired *t* test was used to compare players in the RTP group with the matched control group for statistics before and after the injury season. Statistics were generated using SPSS.

Results

We searched rosters of all 32 NFL teams from the 2009-2016 seasons, including the preseason players (Figure 1). Of the 63 players included in our analysis, 36 achieved RTP (57.1%) while 27 players did not RTP (42.9%) (dnRTP). A comparison of the offensive and defensive players who achieved successful RTP is shown in Tables 1 and 2, respectively. The TE and LB groups had the highest RTP rate at 71% and 70%, respectively, while the WR and RB groups had the lowest RTP at 38% and 40%, respectively. The DB group had a RTP rate of 62%. With the numbers available, no significant difference could be detected between the mean age at the time of injury between the RTP and dnRTP groups. On average, players who achieved RTP had more career games and percent games started than players who did not achieve RTP. The offensive RTP group had an average of 78.33 career games played prior to the injury compared with the dnRTP group, which had 25.44 career games played (P = .0072). Similarly, the defensive RTP group had an average of 93.63 career games played prior to the injury compared with 33.09 for the dnRTP group (P = .0043). With regard to the percent games started, the offensive RTP group started an average of 43 games (55.8%) compared with the dnRTP group, which started an average of 2 games (9.33%) (P = .0213). The defensive RTP group started an average of 63 games (67.95%), while the dnRTP group started only an average of 6 games (18.82%) (P = .0151). Seven players did not have preinjury statistics to report and were not included in the pre- and postinjury performance comparison in Table 3. Two players suffered repeat AT ruptures and were only included in the RTP group comparison for the primary injury and were excluded from further statistical analysis.

TEs achieved successful RTP at a faster rate than other position groups. The TE group time to RTP averaged 11.25 months compared with the RB, DB, WR, and LB groups where RTP was achieved at 11.75, 11.91, 12.4, and 12.63 months, respectively, with the LB group taking the longest time to RTP. The average RTP for the entire cohort was 12.6 months. There was no statistical significance in RTP times for the position groups.

With regard to in-game preinjury performance (Table 1), WRs and TEs in the RTP had significantly more yards/reception (P = .0021), receptions/game (P = .0026), and receiving TDs/season (P = .0163) than the players in the dnRTP group. RBs in the RTP had more yards/rush, rushes/game, and rushing TDs/season compared with the RBs in the dnRTP group. For the defensive players, the RTP group had better preinjury performance than the dnRTP group (Table 2) in all performance categories measured; however, none reached statistical significance. With the exception of passes defended, players in the RTP group did not demonstrate any appreciable difference when comparing their pre- to postinjury individual statistics. Compared to preinjury statistics, players from all positions in the RTP group showed a significant decrease in postinjury performance in all statistical categories calculated (Table 3).

Compared with the RTP group, WRs and TEs in the matched control group had more post injury receptions per game, more yards per reception, and more TDs per season, however only the difference in receptions per game was statistically significant. When comparing the defensive RTP group with the control group, the control group had better pre- and postinjury statistics in all categories (Table 3). Statistical significance was achieved in tackles (P = .0069), passes defended (P = .452), and fumbles forced/recovered (P = .0751) (Table 3).

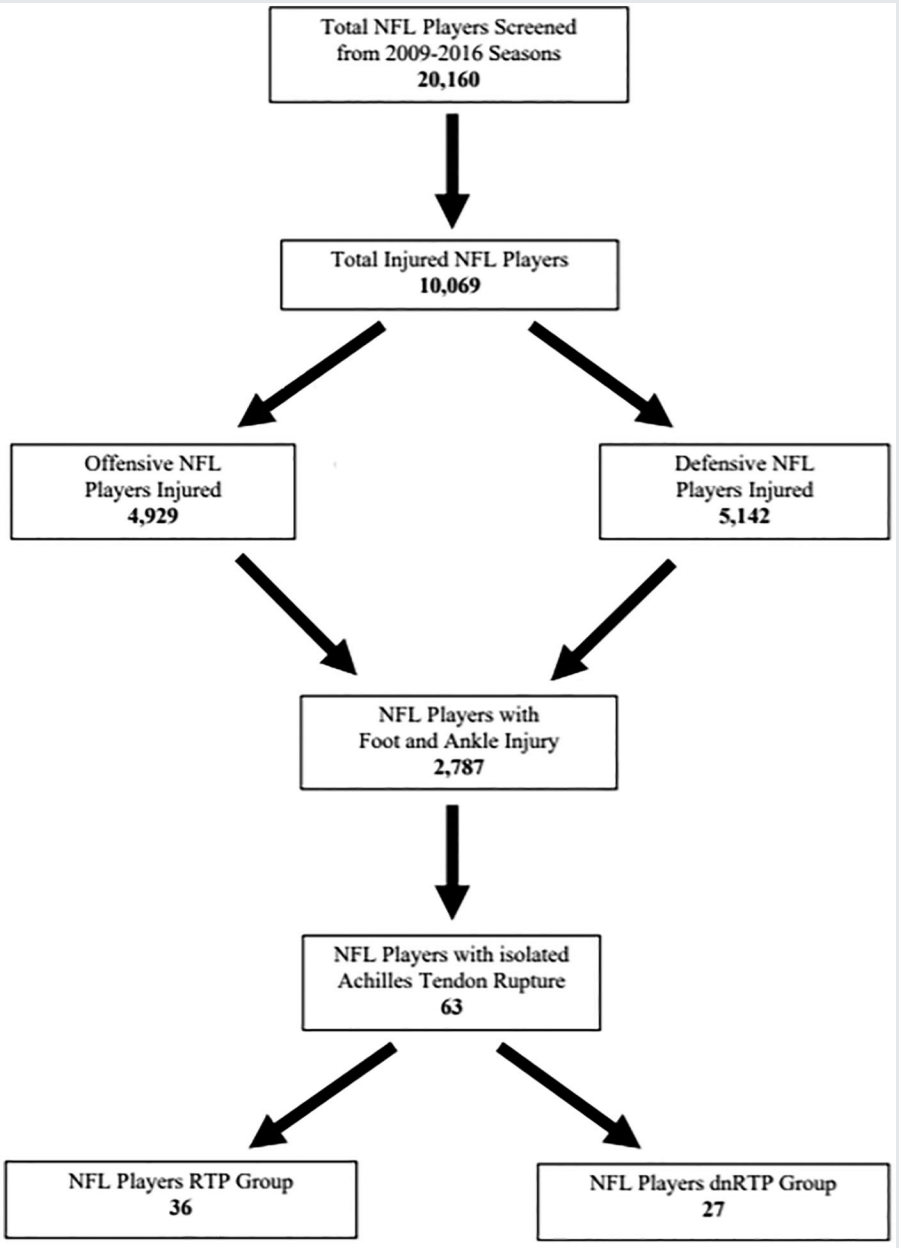
Discussion

The primary goal of this study was to report on and compare the RTP rates in specific NFL offensive and defensive skill positions (Table 4) from the 2009-2010 through 2015-2016 seasons and compare pre- and postinjury player performance. Our study has the lowest reported RTP rate of 58% when compared with previous studies that found RTP rates of 61.3%, 65.6%, 72.4%, and 72.5%.[22,24-26] Previous studies have referenced a preexisting NFL injury database constructed primarily for research while our data collection was unique as each team roster was searched to identify injured players.[22,26] Our study offers a purposeful approach for the control group as we found matched control players for each injured player. We felt that combing through rosters would allow us to (*a*) not miss any AT injuries and (*b*) be able to appropriately identify control players while providing the most performance detail of any sports-related AT study in the current literature. We chose to report true performance measures to provide the actual impact that AT rupture has on player performance. Previous reports have either not reported game performance, used surrogate measures of performance, including subjective scoring systems or reported on games played and games started as a measure of injury impact on postinjury performance.[22,24,26] While important to clarify, these metrics do not substitute for the reporting and discussion of true player in-game performance. Furthermore, the use of a subjective scoring system does not translate directly to performance statistics and sports vernacular.

Consistent with prior data, the RB group had a low RTP rate of 40%. This is

Figure 1.

Flowchart demonstrating the inclusion process for this study, ultimately demonstrating the return to play (RTP) and did not return to play (dnRTP) group numbers.



likely multifactorial as the RB position has a high turnover rate in the NFL, with an average career length of only 2.5 years.[29] As NFL offenses have evolved, the role of RBs has changed dramatically with many teams utilizing several RBs instead of relying on just one.[29-32] Additionally, RBs rely on fine-tuned footwork in order to move quickly in tight quarters with rapid and explosive change of pace and direction, often colliding with much larger players. WRs had the lowest RTP at 38%. Our finding is similar with the 40% reported by Yang et al.[22] The skills required to play WR are similar to those mentioned for the RB group, especially the swift change of speed and direction. Even after a successful primary AT repair, if the WR has lost even a portion of his ability to accelerate, his job is in jeopardy.[33] A paradoxical result was the 62.5% RTP rate for DBs. They require a similar skillset as WRs; however, they returned

Table 1.

Demographics and Performance Before Injury for Offensive Skill Players.[a]

	RTP (n = 14)	Did not RTP (n = 16)	*P*
Demographics			
Age at time of injury, years	26.14 ± 3.88	25.81 ± 3.03	.7958
Timing of injury (n)			
Preseason	4	3	
Regular season	7	4	
Playoffs	1	0	
Offseason	2	9	
Position (%)			
RB	40.0	60.0	
WR	38.5	61.5	
TE	71.4	28.6	
Performance			
Career games played (n)	78.33 ± 62.79	25.44 ± 36.36	**.0072**
Career starts (% of games)	55.80 ± 64.14	9.33 ± 13.65	**.0213**
WR/TE			
Yards/reception	11.17 ± 4.78	4.75 ± 5.84	**.0021**
Receptions/game	2.33 ± 1.93	0.26 ± 0.3.89	**.0026**
Receiving TDs/season	2.05 ± 2.44	0.10 ± 0.32	**.0163**
RB			
Yards/rush	4.26 ± 1.00	3.35 ± 2.11	—
Rushes/game	12.96 ± 5.94	4.39 ± 5.14	—
Rushing TDs/season	3.80 ± 2.95	1.54 ± 4.12	—
Time to return (months)			
RB	11.75 ± 1.5		.7961
WR	12.4 ± 7.76		
TE	11.25 ± 6.86		

Abbreviations: RTP, return to play; RB, running back; WR, wide receiver; TE, tight end; TD, touchdown.
[a]Data are recorded as mean ± standard deviation, unless otherwise indicated. RTP was defined as playing in at least 1 NFL game after Achilles tendon repair. Performance in this table includes in-game statistics for the 2 seasons preceding the injury.
The bold entries signify statistically significant values where $p < 0.05$

Table 2.

Demographics and Performance Before Injury for Defensive Skill Players.[a]

	RTP (n = 22)	Did not RTP (n = 11)	*P*
Demographics			
Age at time of injury, years	27.11 ± 2.90	25.55 ± 3.33	.19
Timing of injury (n)			
Preseason	5	3	
Regular season	13	4	
Playoffs	0	0	
Offseason	4	4	
Position (%)			
DB	62.5	37.5	
LB	70.5	29.5	
Performance			
Career games played	93.63 ± 56.35	33.09 ± 41.32	**.0043**
Career starts (% of games)	67.95 ± 57.26	18.82 ± 33.58	**.0151**
Tackles	39.4 ± 39.99	21.63 ± 30.83	.089
Tackles for loss	2.82 ± 3.86	1.05 ± 1.65	.06
Sacks	1.26 ± 2.43	0.316 ± 0.73	.104
Passes defended	3.36 ± 5.28	2.05 ± 3.39	.326
Interceptions	0.75 ± 1.28	0.68 ± 1.2	.849
Fumbles forced and recovered	0.93 ± 1.3	0.37 ± 0.68	.08
Total defensive touchdowns	0.16 ± 0.53	0 ± 0	.194
Time to return, months			
DB	11.91 ± 1.04		.2191
LB	12.63 ± 1.41		

Abbreviations: RTP, return to play; DB, defensive back; LB, linebacker.
[a]Data are recorded as mean ± standard deviation, unless otherwise indicated. RTP was defined as playing in at least 1 NFL game after Achilles tendon repair. Performance in this table includes in-game statistics for the 2 seasons preceding the injury.
The bold entries signify statistically significant values where $p < 0.05$

at a much higher rate. Differing offensive and defensive schemes and a player's specific role in those schemes may explain this discrepancy. The TE and LB groups had the highest RTP rates at 71% and 70%, respectively. Both of these positions are traditionally played by larger athletes who do not rely on cutting and direction change as do the some of the other skill positions.

We found a significant association between the offensive and defensive RTP groups and the number of career games played and the percent of games started, consistent with previous work,

Table 3.

Performance Before AT Rupture and After Reconstruction.[a]

	Group with successful RTP (n = 27)			Control group (n = 27)		
	Before	After	*P*	Before	After	*P*
WR/TE						
Yards/reception	11.41 ± 4.93	9.87 ± 4.52	.3412	12.64 ± 2.50	11.68 ± 3.53	.2135
Receptions/game	2.60 ± 1.86	2.33 ± 2.36	.5553	4.57 ± 2.38	4.49 ± 2.12	**.0106**
Receiving TDs/season	2.79 ± 2.58	3.64 ± 4.47	.4857	5.50 ± 4.05	5.06 ± 3.44	.3195
RB						
Yards/rush	4.26 ± 1.00	2.37 ± 2.30		3.70 ± 0.47	3.43 ± 0.96	—
Rushes/game	12.96 ± 5.94	3.47 ± 3.02		9.5 ± 7.75	10.28 ± 7.28	—
Rushing TDs/season	3.8 ± 2.95	1.00 ± 1.73		4.17 ± 2.86	2.75 ± 0.96	—
LB/DB						
Tackles	64.54 ± 37.74	51.13 ± 38.64	.2122	70.47 ± 31.55	70.19 ± 27.35	**.0069**
Tackles for loss	4.75 ±4.30	3.25 ± 3.69	.1080	4.91 ± 4.77	3.94 ± 3.98	.2982
Sacks	1.96 ± 2.85	1.75 ± 2.99	.6202	2.22 ± 3.49	1.92± 2.66	.6129
Passes defended	5.04 ± 5.85	2.79 ± 4.27	**.0306**	5.53 ± 5.27	4.50 ± 3.45	**.0452**
Interceptions	1.26 ± 1.48	0.78 ± 1.35	.2173	1.38 ± 1.48	1.22 ± 1.48	.1585
Fumbles forces/recovered	1.38 ± 1.41	0.79 ± 1.38	.1661	1.75 ± 1.87	1.41 ± 1.60	**.0751**
Defensive TDs	0.29 ± 0.69	0.17 ± 0.38	.4786	0.28 ± 0.58	0.19 ± 0.47	.7242

Abbreviations: AT, Achilles tendon; RTP, return to play; WR, wide receiver; TE, tight end; TD, touchdown; RB, runing back; LB, linebacker; DB, defensive back.

[a]Data are recorded as mean ± standard deviation. For each combination of matched players, the before data includes two seasons preceding the injury and the injury season. The after data includes two seasons the injury season. RTP was defined as playing in at least 1 NFL game after AT repair.

The bold entries signify statistically significant values where $p < 0.05$

which has shown a correlation between draft pick position and likelihood of returning after injuries.[22,34] The difference in games played and percent games started between the 2 groups was significant and the player's overall talent and abilities likely played an important role. One potential explanation is the role of star players. If a star player is injured and then required to sit out for a substantial period of time, the franchise is more likely to keep this player on the roster during the recovery period compared to a player with a more limited role. As a result, players who did not RTP, may have failed for a variety of factors unrelated to AT rupture and recovery, including individual talent and specific team needs. There may be a correlation with tendon degeneration and decrease in ability such that these players were allowed fewer playing opportunities.[10,35,36]

The average time to RTP for all the included positions was 12.6 months while previous studies have reported RTP between 11 and 12 months.[22,24,27] Our data show that RBs were able to return earlier than the other positions, at 11.75 months. TEs took the longest to return at 14.33 months. Importantly, no player was able to achieve RTP within the same season consistent with prior data that time to RTP is longer than the length of a season.[21,22,37,38] Maffulli et al[9] reported a mean RTP time of 4.8 months for elite athletes; however, the study did not include professional football players. Currently, we are not aware of any reports, operative techniques, or accelerated rehabilitation programs that allow a player to RTP within the same season.

Table 4.

Role of Each Position Reported and Required Skills.

Position	Role/skills
RB[a]	Carry the ball, evade tacklers, catch passes, block larger players. Precise footwork, maneuver in tight spaces, speed.
WR	Catch passes, block defenders. Speed, change of pace, deception.
TE[a]	Block large players, catch passes. Body control, strength.
DB	Defend WRs, tackle larger players. Extreme change of speed and direction, quick reaction times.
LB[a]	Evade larger blockers, tackle runners and receivers, defend passes. Lateral mobility, speed, strength, field vision.

Abbreviations: RB, running back; WR, wide receiver; TE, tight end; DB, defensive back; LB, linebacker.
[a]Frequent high-impact collisions.

Interestingly, players who were injured in the offseason and preseason periods were less likely to achieve RTP. NFL rosters are larger during those periods while the teams are trying to choose players in order to field the best possible roster for the upcoming season. Players competing for roster positions are put through grueling work outs and often given more repetitions than the established or star players. This increase in playing opportunity could potentially translate to an increase in career-ending injuries in position players who are already in danger of not making the team.

When comparing the RTP and dnRTP groups, the RTP groups for both offensive and defensive players had better pre-injury game performance measures. This supports the previously established idea that players who are able to achieve RTP were better players to begin with compared to the dnRTP group.[22,34] Some position groups improved their performance after AT repair. For example, the defensive players had statistically significant increases in passes defended. The WRs and TEs had more receiving TDs per season; however, the difference did not reach statistical significance. The comparison of the RTP group to the control players found that the controls outperformed the RTP group in all categories for every position group for both pre- and postinjury periods. A combination of the control group being composed of better players as well as the having an extra season to improve while the RTP group was injured may explain this finding.

This represents the only known study to describe postinjury in-game performance for specific statistics (eg, yards per reception, yards per carry, etc). Other studies have reported on postinjury performance but not in a fashion that shows true performance measures.[22,27] To our knowledge, players included in the study had isolated primary AT rupture. To avoid confounding factors, players with rerupture or contralateral rupture were excluded.

Financial implications drive many decisions for NFL organizations and players. Player contracts and amount of guaranteed and nonguaranteed money directly affects a team's ability to keep or cut players who are injured or who have a history of injury. With few roster spots available in a highly competitive league, players with AT rupture vacate a roster spot allowing other players to improve and possibly permanently replace them.

There are a number of limitations to this study. It is a retrospective review of publicly available data, therefore certain variables, such as concomitant injuries, could have been underreported and therefore missed. The means of identifying injured players was challenging, as the media reported more heavily on star players. Players who were only on the preseason roster likely did not garner the same amount of media attention and detailed reporting as some high-profile players. In addition, in some cases, we were not able to determine if players with AT rupture had suffered from associated AT pathology such as tendinopathy, sprains, and so on prior to their injury. As a whole, the sample size is relatively small and thus prohibits a powerful analysis specifically of performance data. Because of small sample size we were unable to make statistical comparisons for the RB group. Because there was no intervention, the control group serves more as a frame of reference than a traditional control group. We aimed to create a group with similar opportunities to compare against. Usually there are more than one player of each position type on the field at the same time, with the exception of RB, making this a difficult comparison. While the treatment in NFL players for AT rupture is primary repair, there are a variety of approaches and techniques that may affect the outcome. The information about the specific operative technique employed in each scenario was not readily available. Additionally, clinical follow-up and rehabilitation protocols were unknown. Reasons for dnRTP are not fully known, however could also include player age, open positions on team rosters, and decreased ability. This study was limited to NFL RBs, WRs, TEs, DBs, and LBs and may

not be applicable to other position groups or sports.

Conclusion

Our study found that 57% of NFL skill position players achieved RTP with WRs and RBs having the lowest RTP rates and TEs and LBs having the highest RTP rates. RBs, DBs, and LBs with successful RTP had decreased performance in all measured categories. This updated information may be helpful for competitive athletes, team physicians who care for and counsel athletes about the likelihood of RTP and associated recovery time, as well as NFL general managers, scouts, and coaches in evaluating players with a history of AT rupture.

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Ethical Approval

Not applicable, because this article does not contain any studies with human or animal subjects.

Informed Consent

Not applicable, because this article does not contain any studies with human or animal subjects.

Trial Registration

Not applicable, because this article does not contain any clinical trials.

ORCID iDs

Dane Barton https://orcid.org/0000-0001-7954-2133
Ansab Khwaja https://orcid.org/0000-0002-2250-8310 FAS

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